MATH0023 Algebraic Topology

Year:	2026–2027
Code:	MATH0023
Level:	7 (UG), 7 (PG)
Normal student group(s):	UG: Y3, Y4 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	100% examination
Normal Pre-requisites:	MATH0014 (Algebra 3: Further Linear Algebra)
Lecturer:	Prof F Johnson

Course Description and Objectives

The purpose of this course is to provide an elementary introduction to the methods of Algebraic and Geometric Topology via the homology of simplicial complexes. A good grounding in Linear Algebra is assumed, but with that exception the course is self contained.

Detailed Syllabus

Chain complexes and homological simplicial complexes and simplicial homology. Mayer-Vietoris Theorem. Computation of some low dimensional examples $H_*(\Delta^n)$, $H_*(S^n)$, $n \leq 2$. The cone construction. $H_*(S^n)$, $H_*(M_1 \# M_2)$ for simplicial surfaces M_1, M_2. Geometrical realizations. Invariance of $H_*(\ )$ under subdivision. Homology of products. Künneth Theorem.

Classification of closed, connected piecewise linear surfaces as connected sums thus:

$$\mathbb{T}^2\#...\#\mathbb{T}2, \quad \mathbb{RP}^2\#...\#\mathbb{RP}^2, \quad \mathbb{RP}^2\#\mathbb{RP}^2\#\mathbb{RP}^2\# \cong \mathbb{T}^2\#\mathbb{RP}^2$$

Fixed Point Theorems: in particular, Lefschetz complexes and Brouwer Fixed Point Theorems.

August 2026